Amrit Srivastava

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Education

Michigan State University GPA: 3.7

Bachelor of Science in Computer Science, Minor in Data Science *Aug 2022 – May 2026*

Coursework: Object Oriented Design, Algorithm Engineering, Compilers

Experiences: 3x Hackathon Winner, Engineer MSU Rocketry

Experience

Student Library Assistant | *Digital Multimedia Center MSU Library* Jan 2023 – Present

- Automated library procurement by using Python and Selenium to source item prices, saving 2 hours fortnightly
- Enhanced the library games catalog by enriching metadata, enabling advanced sorting and improving patron experience
- Developed Rust application to sort spreadsheets by item call number, addressing limitations of Excel-based system
- Coordinated front-desk operations; facilitated room and media checkouts, maintained a 10,000+ item catalog ensuring resource accessibility and high patron satisfaction

AI Robotics Research Assistant | *Action Lab* June 2025 – Jan 2026

- Evaluated the safety and performance of recently published robot navigation algorithms by developing a realistic city-scale environment in NVIDIA Isaac Sim with high-density vehicle and pedestrian traffic
- Developed ROS2 middleware that connected simulated robots to external computer vision algorithms, allowing the team to "plug-and-play" research algorithms without rebuilding the simulation, reducing compute time
- Implemented high-performance C++ algorithms for robotics, computer vision and control logic
- Authored detailed documentation with annotations and pictures to accelerate team adoption

Data Science Intern | *iSON Xperiences Ltd.* July 2023 – Aug 2023

- Conducted an EDA and trained a machine learning model on a 1.85 million data-point dataset to identify potential paying customers with 87% accuracy, enabling prioritized call center outreach and significantly reduced costs
- Architected and deployed a secure data pipeline with Selenium web scraping; used Python to script and automate ERP data ingestion, saving 1.5 hours/week and reducing human error

Projects

Machine Performance Prediction | *AkzoNobel Capstone Project*

Python, React, Machine Learning

- Reduced unplanned machine downtime and missed delivery orders for factory clients by developing a predictive maintenance ML model using Python, Polars, and SciPy to identify equipment abnormalities from real-time sensor data
- Engineered a cross-platform Next.js progressive web application that centralizes sensor data and breakdown predictions

OpenFiche | *Spartahack 2025 (Winner - Specialized Mastery)* www.reada.wiki

Python, React, Natural Language Processing, Algorithm Engineering

- Built and deployed a search engine that surfaces human-authored content by leveraging natural language processing and adapting Google's PageRank search algorithm to demote low-quality autogenerated pages

Rocket Communication System | *MSU Rocketry Team*

C++, Low Latency, Linux, System Architecture

- Developed a multi-threaded C++ low-latency communication protocol for rocket-to-mission-control telemetry exchange
- Architected a self-healing Linux environment via systemd supervision, implementing automated recovery logic and watchdog timers ensuring high availability during flight

Technical Skills

Languages & Frameworks: Python, C++, Rust, Swift, JavaScript, React, SQL, ROS2, React Native

Skills: Process Automation, Software Development, Web Technologies & Automation, Artificial Intelligence, Machine Learning, Computer Vision, Natural Language Processing, Linux Systems, Robotics